Soldering wire:

This is made up of an alloy of lead and Tin which is available in the form of wire in size of 16 to 18 SWG. The percentage of tin and lead is 60% and 40% respectively. Little amount of Paste is inserted internally in the solder wire. Solder wire without paste inside, is also available.

Safety Precautions

1. Plug in the Soldering iron into the relevant plug point and wait for few minutes to heat up.
2. Keep the Soldering iron always in its stand while not in use.
3. Keep a small piece of wet sponge. This is used to clean the tip of the iron and also helps you to do first aid in case of heat wound happens on the finger.
4. Touch the tip of the iron with soldering wire, see that it melts, to check the iron sufficiently heated.
5. Don't touch the tip of the soldering iron, as this heats upto 400˚C.
6. Bit of smoke will come out during soldering. Don't inhale the smoke.
7. Wash your hands after soldering work finished, because the soldering material contains some harmful materials.

Soldering procedure:

1. The melting point of the solder wire should be about 60° C.
2. The solder must be well melted with good flow, so that it can fill up the gaps between the joints.
3. A solder must have sufficient corrosion resistance, so that the joint may not be open due to corrosion.
4. The bit of the iron should always be kept neat and clean.
5. Due to the oxide formation the tip of the soldering iron becomes black and dirty. So the tip surface should be filed and cleaned before soldering is started.
6. The surface of the wire to be soldered should be clean.
7. A joint should be secured by removing the insulation on the wires and twisting them together before soldering.
8. The excessive paste and solder should not be used for joint because it may cause dry soldering.
9. When soldering in a congested place, care must be taken such that, the insulation of wires and other components are not burnt.
10. While soldering ICs, care should be taken that excessive heat does not pass through the leads and pins of IC should not get short.

Result:

The functions of soldering iron, different types of soldering iron, its working function are learnt.

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